

Technology Stack (Architecture & Stack)

Date: 03/10/2026	Team ID: 6ab8f707c23107aaaa73276f
Project Name: Auto Ticket Classification using Flow Designer	Maximum Marks: 4 Marks

1. Technical Architecture Overview

Architecture Flow: [End-User / Caller via Service Portal / Email] →[ServiceNow Cloud Instance (sys_user link)] →[Incident Workflow Record Created (Category == EMPTY)] →[Flow Designer Trigger Engine (sys_id)] →[Action: Check Short Description String contains Keywords (Wi-Fi, Projector, Password, Slow PC)] →[Action: Update Record Category & Dynamic Dependent Subcategory] →[Action: Send Automated Caller Confirmation Email] →[Resolver Assignment Queue / Dashboard Monitoring UI]

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Portal & workspace where callers log tickets and agents view categorized queues	ServiceNow Service Portal, Next Experience Workspace, HTML5/Angular
2	Application Logic-1	Automated incident event trigger and execution guardrails (Category is empty)	ServiceNow Flow Designer Trigger Engine (Asynchronous Execution)
3	Application Logic-2	Keyword condition parser evaluating Short Description text string	Flow Designer Native Conditional Logic (IF/ELSE IF Branches)
4	Application Logic-3	Automated caller notification dispatch action dynamically pulling sys_user profile	Flow Designer Outbound Email Action (ServiceNow Notification Engine)
5	Database	Incident data storage, dependent choice fields, and reference pointers	ServiceNow Relational Schema (MariaDB/MySQL-based backend)
6	Cloud Database	Enterprise multi-tenant cloud storage hosting user and incident tables	ServiceNow High-Availability Cloud (Multi-Instance Architecture)
7	File Storage	Storage for project artifacts, execution logs, and XML update sets	ServiceNow Attachment Engine & Project Update Set Repository
8	External API-1	Outbound email transport protocol for transmitting caller receipt emails	ServiceNow SMTP Server / Notification Engine
9	External API-2	Future integration endpoint for campus single sign-on / LDAP user directory	REST API / LDAP Integration Protocol
10	Machine Learning Model	Future predictive keyword extraction and NLP categorization engine	ServiceNow Predictive Intelligence (Classification Framework)
11	Infrastructure (Server/Cloud)	Enterprise SaaS deployment hosting the custom workflow instance	ServiceNow SaaS Cloud Instance (Washington DC / Utah Release)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source / Platform Frameworks	Standard low-code workflow engine executing conditional branching without custom scripts	ServiceNow Flow Designer Framework, ITIL ITSM Foundation

2	Security Implementations	System User context execution, Table ACLs limiting editing to ITIL roles, and flow_designer admin permissions	ServiceNow Access Control Lists (ACLs), Role-Based Access Control (RBAC), TLS 1.3
3	Scalable Architecture	Scales smoothly across 1,200+ monthly school tickets without performance degradation	ServiceNow Multi-Instance Cloud Architecture with asynchronous background queuing
4	Availability	High fault-tolerance and zero-downtime service across paired cloud data centers	Advanced High Availability (AHA) multi-datacenter data replication
5	Performance	Rapid, non-blocking execution averaging ~180 ms per incident classification run	ServiceNow Event Queue Engine & Asynchronous Execution Threading